

Senthil Sengodan

Senior GIS Manager



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Seasoned Senior Manager and Geospatial Solutions Leader with over 15 years of global experience in driving enterprise GIS strategy, digital twin development, and location intelligence solutions across public and private sectors. Proven success in managing end-to-end GIS service delivery and smart infrastructure programs through both onsite and offshore implementations.

Known for leading cross-functional teams, optimizing operations, and aligning geospatial initiatives with organizational goals. Adept at stakeholder engagement, IT service governance, and innovation leadership. Highly proficient in ESRI technologies, ITIL-based frameworks, cloud-native environments (AWS, Kubernetes), and complex FME-based system integrations. Committed to delivering value-driven outcomes and long-term digital transformation.



Work history

2021-03 -
2025-03

Senior Manager – Geospatial Solutions (*Smart City Technology Division*)

Government Technology Agency (GovTech), Singapore

March 2021 – March 2025

- Spearheaded strategic geospatial initiatives for three major government agencies — Singapore Land Authority (SLA), Urban Redevelopment Authority (URA), and Ministry of Manpower (MOM) — delivering data-driven urban solutions, digital twin platforms, and simulation-based decision support systems.
- At SLA (2021–2022), led the Proof-of-Concept deployment of ArcGIS Enterprise on AWS GCC using Kubernetes. Supported ongoing GeoSpace operations, built custom data interfaces for new agency integrations, and facilitated Design Thinking workshops to shape the next-generation Digital Twin GEO (DTGEO) platform.
- At URA (2022–2023), set up a cloud-based high-performance computing environment using AWS to support OpenFOAM CFD simulations, developed FME workbenches for 3D data conversions, and implemented voxel-based web visualizations by integrating ArcGIS with FME Flow.
- At MOM (2022–2023), directed the design of a Chemical Dispersion Solver using OpenFOAM for hazard simulation, developed pre- and post-processing automation tools, and built an interactive geospatial viewer for visualizing scientific results, supporting public safety initiatives.
- From 2023–2025, Led the development of an advanced geospatial platform that automated voxel layer creation in ArcGIS Pro using VTK outputs, enhancing 3D visualization and situational awareness for Digital Twin applications.

- Directed cross-agency projects to deliver spatial decision-support tools and simulation frameworks, enabling smarter urban planning and operational efficiency across SLA, URA, and MOM.
- Championed AI and simulation innovations, including a Chemical Dispersion Solver using OpenFOAM and an in-house free-text geocoder powered by Google BERT for smarter geospatial search and hazard assessment.

2021-03 -
2022-04

Senior IT Manager

Singapore Land Authority (SLA), Singapore

- Led a Proof-of-Concept for deploying ArcGIS Enterprise on Kubernetes in AWS Government Commercial Cloud, validating a scalable geospatial solution and demonstrating ArcGIS Server implementation feasibility in restricted environments
- Provided critical technical support and optimized the National Spatial Data Infrastructure (NSDI) system (GeoSpace), enhancing system reliability and increasing data conversion/transfer efficiency by 20% via FME Server integration
- Facilitated Design Thinking workshops with 10+ agencies and over 50 stakeholders to define requirements for the upcoming Digital Twin Project, ensuring alignment with business and operational needs
- Evaluated hybrid cloud solutions, managed vendor relationships, and prepared cost estimates and tender documentation for \$70M+ smart city initiatives
- Negotiated vendor contracts for hardware, software, and services, securing 10-15% cost savings while ensuring high-quality procurement
- Collaborated with cross-functional teams, integrating IT solutions that strengthened system security and streamlined workflows, reducing manual efforts by 25%

2014-06 -
2021-03

GIS Lead Consultant / Project Manager

NCS PTE LTD, Singapore

- Managed and enhanced the SLA Cadastral Survey System, supporting national land survey operations
- Led implementation of a 3D Digital Twin for the HDB SMART Hub, enabling advanced urban analytics
- Delivered a real-time monitoring system for the LTA (Land Transport Authority) improving transportation network visibility
- Architected and oversaw the unification of disparate GIS systems into a centralized enterprise platform for SLA, enhancing data accessibility and reliability
- Designed and managed enterprise GIS architectures, ensuring scalability and performance for government clients
- Provided pre-sales technical leadership, including developing POCs for 3D modeling, geospatial analytics, and big data visualization (Tableau, Qlik, Power BI)
- Developed an automated 3D modeling engine for generating building models (BIM, IFC, CityGML) to support advanced geospatial applications
- Developed custom solutions based upon clients' strict requirements

- Prepared project lifecycle documentation and project management plans

2010-07 -
2014-12

GIS Architect / Project Manager

TechMahindra Singapore PTE LTD, Singapore

- Led the architecture, development, and implementation of GeoSpace, Singapore's National Spatial Data Infrastructure (NSDI) platform for SLA, overseeing a multi-million dollar project using ESRI Enterprise Solutions with FME integration
- Architected and deployed large-scale enterprise GIS systems using ESRI technologies, enabling seamless national data integration
- Managed and mentored GIS development teams throughout the full project lifecycle, ensuring successful delivery within deadlines and budgets
- Designed and implemented raster-based Slope Analysis using ArcGIS Engine, developing an innovative raster analysis tool for heatmap generation and point density analysis using ESRI Flex API
- Developed optimized workflows for geospatial data handling using Oracle Spatial and ESRI ArcSDE, significantly improving system performance and scalability
- Collaborated closely with multiple government agencies to align project goals with national spatial data standards

2009-01 -
2010-07

Senior GIS Specialist

Fugro Data Solutions PTE LTD, Singapore

- Designed and delivered ArcGIS Enterprise solutions for offshore/maritime industries, optimizing vessel tracking and pipeline monitoring, reducing operational risks by 30%
- Developed an Automated Vessel Tracking System (AVTS) integrating ArcGIS Server web APIs with real-time GNSS data, achieving a 20% improvement in vessel tracking accuracy
- Architected and implemented a solution for monitoring subsea gas pipeline integrity (risk management, erosion estimation, APDM integration), reducing inspection time by 25%
- Contributed to the centralized GIS portal and ArcGIS database, managing data for over 100 job sites

2008-04 -
2008-10

Senior Technology Consultant

GeoData AS (ESRI Distributor Norway), Norway

- Designed and implemented a Smart Metering System for Gothenburg Energy (Sweden), automating readings for 270,000+ households using ArcGIS Server and ZigBee, reducing manual effort by 80%
- Customized ArcGIS Server with .NET to integrate geospatial data into the AiMiR Advanced Meter Management system, reducing errors by 25%
- Integrated ZigBee wireless networks for seamless communication between smart meters and the central system, improving data transmission reliability by 30%

2005-12 -
2008-04

Team Lead / Sr. GIS Developer

NIIT GIS Ltd (ESRI Distributor), New Delhi, India

- Led development of ArcGIS Server web applications for telecom network planning, improving planning efficiency by 25%
- Customized ArcGIS Desktop to integrate SCADA within electricity network management systems (AM/FM) for real-time monitoring across 200+ substations
- Managed complex DGN to ArcSDE data conversion (over 2TB) for the Indian Army's geospatial system
- Engineered network tracing algorithms (upstream/downstream) using VBA/VB.NET, reducing processing time by 40%
- Directed development teams (up to 20 members), conducted code reviews, and ensured timely delivery with a 98% client satisfaction rate



Education

2003-06 - 2004-05

Research/Project Associate: Environmental And Water Resources Engineering

Indian Institute of Technology Madras (IITM) - Chennai

R&D Project: GIS-Based Risk Assessment and Mathematical Modelling for Urban Piped Water Supply System using ArcView Avenue scripting and VB

2001-06 - 2003-03

Master of Engineering: Environmental Applications And GIS

ANNA UNIVERSITY College of Engineering Guindy - Chennai, INDIA

1996-06 - 2000-04

Bachelor of Engineering: Mechanical Engineering

Bharathiar University - Coimbatore, INDIA



Tools And Technologies

ESRI ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, FME Server, AWS, Kubernetes, Azure (familiarity), Python, OpenFOAM, CesiumJS, Tableau, Leaflet, Power BI, Oracle Spatial, AutoCAD, Revit, BIM, CityGML



Professional Recognition

- Delivered GeoSpace NSDI Platform, Singapore's national geospatial system.
- Presented research at ISRS on GIS-based urban water distribution modelling.
- Supervised teams of 20+ in full-cycle GIS solution development.



Personal Information

Work Permit: H1B (SAIPSIT Inc)